PRISMA-Infant-Outcomes (Issued: 2026-05-06)

Includes data from synapse last updated: 2026-05-01

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1. Summary

Table 1. Summary of all infant outcomes included in this report.

Denominator: All livebirths unless otherwise specified. All missing is excluded.

Table 1

	Total	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Total deliveries [a]	16001	2420	2257	1949	2441	5323	1611
Total livebirths	15581	2363	2212	1906	2372	5176	1552
Low birthweight <2500g (PRISMA or facility measured), n (%)	2953 (19)	326 (14)	353 (16)	476 (25)	266 (11)	1279 (27)	253 (17)
Low birthweight <1500g (PRISMA or facility measured), n (%)	183 (1.2)	22 (0.96)	25 (1.1)	21 (1.1)	12 (0.51)	65 (1.4)	38 (2.6)
Preterm birth <37wks, n (%)	2268 (15)	159 (6.7)	190 (8.6)	301 (16)	205 (8.6)	1211 (23)	202 (13)
SGA <3rd, n (%)	1601 (10)	262 (11)	174 (7.9)	261 (14)	172 (7.4)	580 (12)	152 (10)
SGA <10th, n (%)	3963 (25)	661 (29)	493 (22)	631 (33)	425 (18)	1371 (30)	382 (26)
LGA >=90th, n (%)	326 (2.1)	50 (2.2)	45 (2)	20 (1.1)	80 (3.4)	90 (1.9)	41 (2.8)
Neonatal mortality, n per 1000[b]	349 (23)	27 (11)	10 (4.6)	37 (19)	23 (9.7)	211 (41)	41 (26)
Infant mortality, n per 1000[c]	411 (38)	31 (17)	16 (13)	39 (35)	36 (20)	239 (72)	50 (34)
Stillbirth >=20wks, n per 1000[d]	420 (26)	57 (24)	45 (20)	43 (22)	69 (28)	147 (28)	59 (37)
Stillbirth >=28wks, n per 1000[d]	245 (15)	36 (15)	11 (4.9)	29 (15)	39 (16)	99 (19)	31 (19)
Total fetal deaths, n (%)[e]	667 (4.1)	99 (4)	70 (3.1)	87 (4.4)	103 (4.2)	236 (4.4)	72 (4.4)
Perinatal birth asphyxia, n (%)	1403 (9)	255 (11)	154 (7)	216 (11)	115 (4.8)	535 (10)	128 (8.2)
Hyperbilirubinemia by AAP threshold, n (%)[e]	549 (3.7)	51 (2.3)	92 (4.4)	87 (4.6)	58 (2.5)	230 (4.6)	31 (2.2)
Any jaundice identified at any time[g]	3503 (23)	171 (7.6)	425 (20)	297 (16)	310 (13)	2145 (43)	155 (11)
PSBI at IPC, n (%)[h]	1347 (8.6)	172 (7.3)	198 (8.9)	185 (9.7)	143 (6)	598 (12)	51 (3.3)

[a] Total deliveries include all livebirths and stillbirths.

[b] Denominator is all livebirths with MNH09 and MNH11 filled out AND have passed the risk period with a visit OR died within the risk period (risk period: age = 28days).

[c] Denominator is all livebirths with MNH09 and MNH11 filled out AND have passed the risk period (risk period: 365days past birth).

[d] Denominator is all livebirths and stillbirths.

[e] Denominator is all infants with a fetal loss reported in MNH04 `(PRG_DSDECOD)` or birth outcome in MNH09 `(BIRTH_DSTERM_INF1)`.

[f] All infants with a TcB measurement taken at <14 weeks of life.

[g] All infants with a jaundice assessment at <14 weeks of life.

[h] Denominator is all livebirths aged 0-59 days.

2. Low birthweight

Definition: Defined as liveborn infant weighing less than 2500g at birth.

Denominator: All livebirths.

To be included as "non-missing" for this outcome, a participant must have:

1. Live birth (varname: `BIRTH_DSTERM [MNH09]`).

2. Birthweight measured by PRISMA staff <72 hours following birth (varnames [form]: `BW_EST_FAORRES [MNH11]`, `BW_FAORRES [MNH11]`, `BW_FAORRES_REPORT [MNH11]`).

3. Facility reported birthweight where PRISMA <72 hours not available (varnames [form]: `BW_FAORRES_REPORT [MNH11]`).

Table 2. Low birthweight

Table 2	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Data completeness [a]						
Denominator	**2363**	**2212**	**1906**	**2372**	**5176**	**1552**
Missing hours since birthweight measured, n (%)	479 (20)	116 (5.2)	160 (8.4)	146 (6.2)	285 (5.5)	214 (14)
Missing PRISMA birthweight, n (%)	328 (14)	340 (15)	324 (17)	148 (6.2)	295 (5.7)	276 (18)
Missing facility birthweight, n (%)	122 (5.2)	7 (0.32)	30 (1.6)	84 (3.5)	1824 (35)	525 (34)
Missing PRISMA & facility birthweight, n (%)	78 (3.3)	7 (0.32)	10 (0.52)	27 (1.1)	91 (1.8)	87 (5.6)
PRISMA measured birthweight [b], n (%)						
Denominator	**2034**	**1871**	**1578**	**2223**	**4207**	**1276**
Normal birthweight >=2500g	1749 (86)	1585 (85)	1220 (77)	1986 (89)	3106 (74)	1079 (85)
Low birthweight 1500 to <2500g	271 (13)	268 (14)	355 (22)	231 (10)	1072 (25)	180 (14)
Very low birthweight <1500g	14 (0.69)	18 (0.96)	3 (0.19)	6 (0.27)	29 (0.69)	17 (1.3)
Any method measured birthweight [c], n (%)						
Denominator	**2285**	**2205**	**1896**	**2347**	**4653**	**1465**
Normal birthweight >=2500g	1959 (86)	1852 (84)	1420 (75)	2080 (89)	3374 (73)	1212 (83)
Low birthweight 1500 to <2500g	304 (13)	328 (15)	455 (24)	254 (11)	1214 (26)	215 (15)
Very low birthweight <1500g	22 (0.96)	25 (1.1)	21 (1.1)	13 (0.55)	65 (1.4)	38 (2.6)

[a] Denominator is all livebirths.

[b] Denominator is all livebirths with a PRISMA-measured birthweight excluding missing.

[c] Denominator is all livebirths with any method measured birthweight excluding missing.

Figure 2a. PRISMA-measured birthweights across sites.

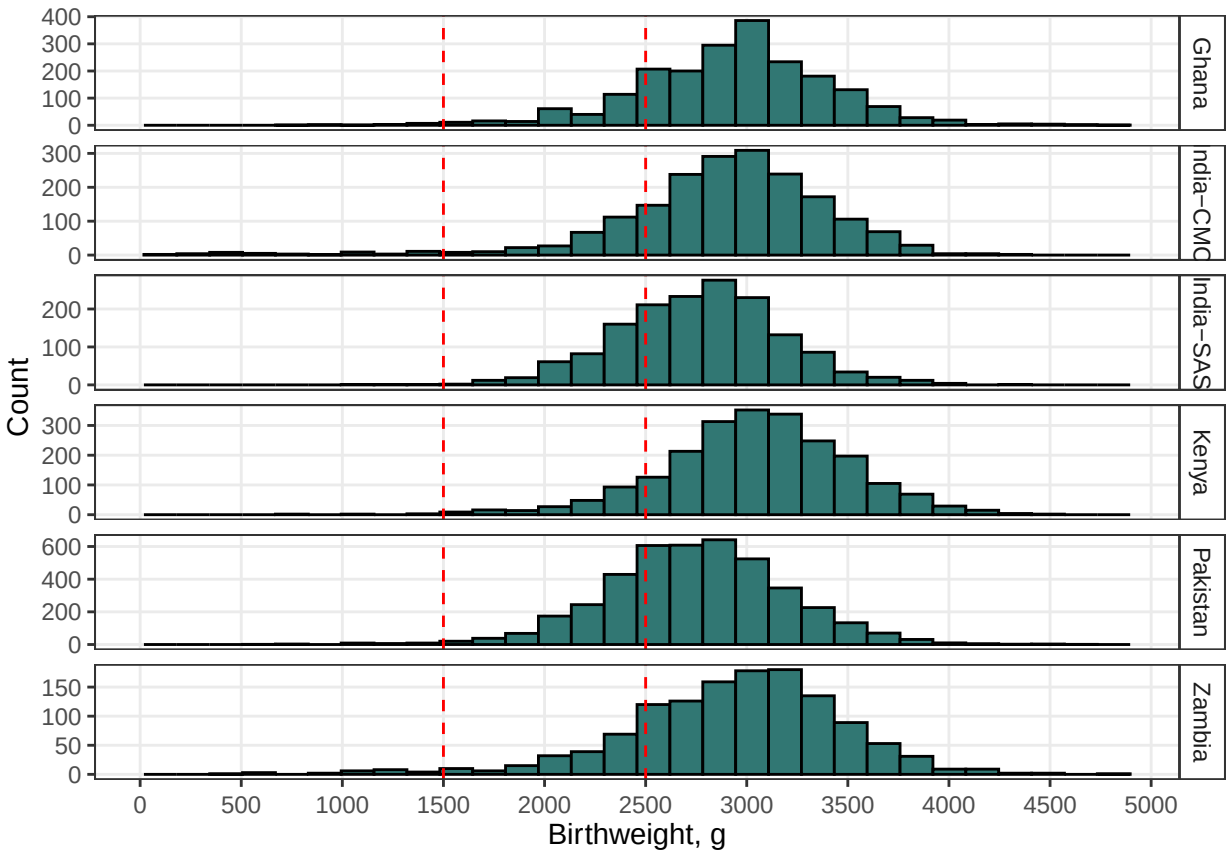


Figure 2b. Facility-reported birthweights across sites.

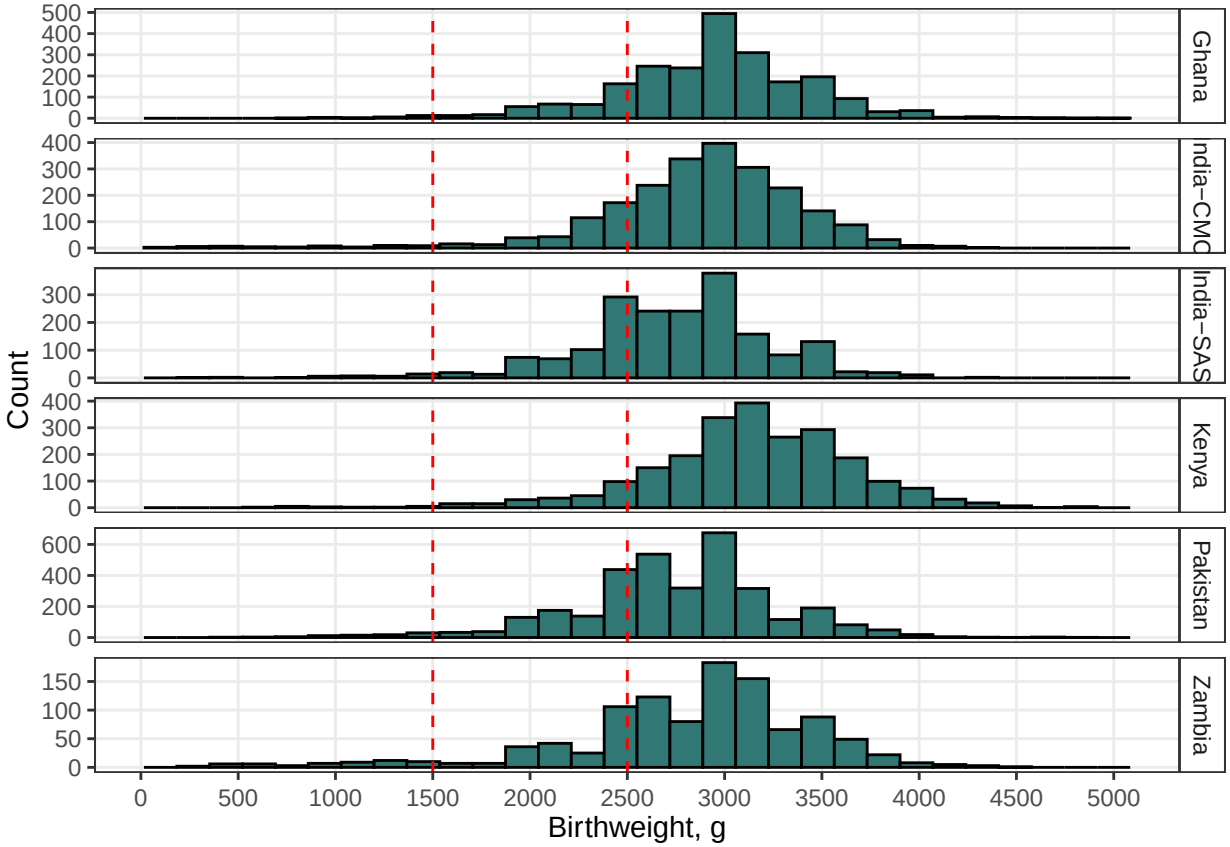


Figure 2c. Hours following birth infant was weighed across sites.

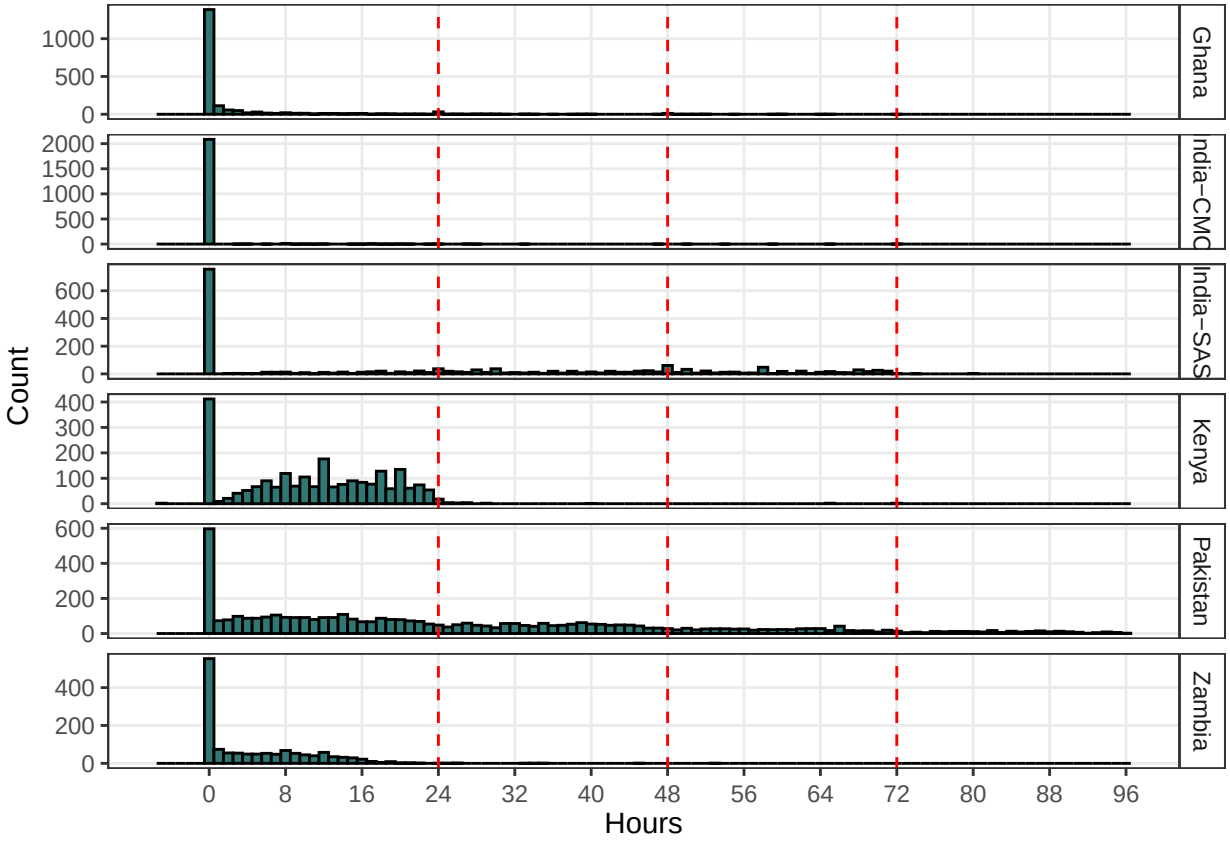


Figure 2c. Hours following birth weight measurement recorded by PRISMA-trained staff (or Facility reported where PRISMA measurement not available). Report of "0" indicates the measurement was taken <1hr following birth. Dashed lines represent 24, 48, and 72 hour time points.

3. Gestational age at delivery

Definition: Preterm delivery prior to 37 completed weeks of gestation. Further classified as: extremely preterm (<28 weeks), very preterm (28 to <32 weeks), moderate preterm (32 to <34 weeks), late preterm (34 to <37 weeks), term (37 to <41 weeks), and postterm (>41 weeks).

Denominator: All participants with delivery after 20 weeks with a MNH09 form filled out with a reported "birth outcome" `(BIRTH_DSTERM_INF1-4=1 OR BIRTH_DSTERM_INF1-4=2 [MNH09])` AND has valid gestational age reported in MNH01 `(US_GA_WKS_AGE_FTS1-4 [MNH01], US_GA_DAYS_AGE_FTS1-4 [MNH01], GA_LMP_WEEKS_SCORRES [MNH01])`. Preterm delivery includes live or stillbirths; preterm birth includes only livebirths.

Note: *Gestational age information collected in MNH01 is used to generate best obstetric estimates for EDD. These constructed variables are then used to calculate GA at time of birth.*

To be included as "non-Missing" for this outcome, a participant must have:

1. Reported gestational age by either LMP or Ultrasound (varnames [form]: `US_GA_WKS_AGE_FTS1-4 [MNH01]`, `US_GA_DAYS_AGE_FTS1-4 [MNH01]`, `GA_LMP_WEEKS_SCORRES [MNH01]`).

2. Valid enrollment ultrasound visit date (varname: `US_OHOSTDAT [MNH01]`)

3. Valid date of birth (varname: `DELIV_DSSTDAT_INF1-4 [MNH09]`).

4. Birth outcome reported as a "Live birth" or "Fetal death" (varname: `BIRTH_DSTERM_INF1-4 [MNH09]`).

Common causes for a participant to be marked as "Missing":

- Participant is missing a reported GA by Ultrasound AND GA by LMP in MNH01.

- Participant has multiple or is missing an enrollment ultrasound visit (`TYPE_VISIT=1`).

- Default value is used for enrollment ultrasound visit date (`US_OHOSTDAT=07-07-1907`).

- Birth outcome is reported as "77, Not applicable".

Table 3. Preterm delivery & preterm birth

Table 3						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Data Completeness [MNH01] (among participants with a reported livebirth or stillbirth)						
Denominator	**2420**	**2257**	**1949**	**2441**	**5323**	**1611**
BOE = Ultrasound, n (%)	2387 (99)	2243 (99)	1915 (98)	1249 (51)	1731 (33)	875 (54)
BOE = LMP, n (%)	33 (1.4)	14 (0.62)	34 (1.7)	1192 (49)	3592 (67)	736 (46)
Preterm birth severity (livebirths) (categorical), n (%)						
Denominator	**2363**	**2212**	**1906**	**2372**	**5176**	**1552**
Extremely preterm (<28wks)	2 (0.08)	5 (0.23)	7 (0.37)	7 (0.3)	33 (0.64)	7 (0.45)
Very preterm (28 to <32wks)	28 (1.2)	15 (0.68)	19 (1)	13 (0.55)	60 (1.2)	28 (1.8)
Moderate preterm (32 to <34wks)	20 (0.85)	19 (0.86)	28 (1.5)	29 (1.2)	127 (2.4)	32 (2.1)
Late preterm (34 to <37wks)	109 (4.6)	151 (6.8)	247 (13)	157 (6.6)	991 (19)	135 (8.7)
Term (37 to <41wks)	1872 (79)	1992 (90)	1551 (81)	1732 (73)	3817 (74)	1163 (75)
Postterm (>=41wks)	332 (14)	30 (1.4)	54 (2.8)	437 (18)	148 (2.9)	187 (12)
Preterm delivery severity (live & stillbirths) (categorical), n (%)						
Denominator	**2420**	**2257**	**1949**	**2441**	**5323**	**1611**
Extremely preterm (<28wks)	23 (0.95)	39 (1.7)	21 (1.1)	37 (1.5)	81 (1.5)	35 (2.2)
Very preterm (28 to <32wks)	32 (1.3)	19 (0.84)	30 (1.5)	18 (0.74)	83 (1.6)	36 (2.2)
Moderate preterm (32 to <34wks)	24 (0.99)	21 (0.93)	31 (1.6)	32 (1.3)	138 (2.6)	37 (2.3)
Late preterm (34 to <37wks)	118 (4.9)	154 (6.8)	251 (13)	166 (6.8)	1022 (19)	143 (8.9)
Term (37 to <41wks)	1888 (78)	1994 (88)	1561 (80)	1747 (72)	3849 (72)	1172 (73)
Postterm (>=41wks)	335 (14)	30 (1.3)	55 (2.8)	444 (18)	150 (2.8)	188 (12)

Figure 3a. Distribution of days difference between GA by US and GA by LMP reporting across sites.

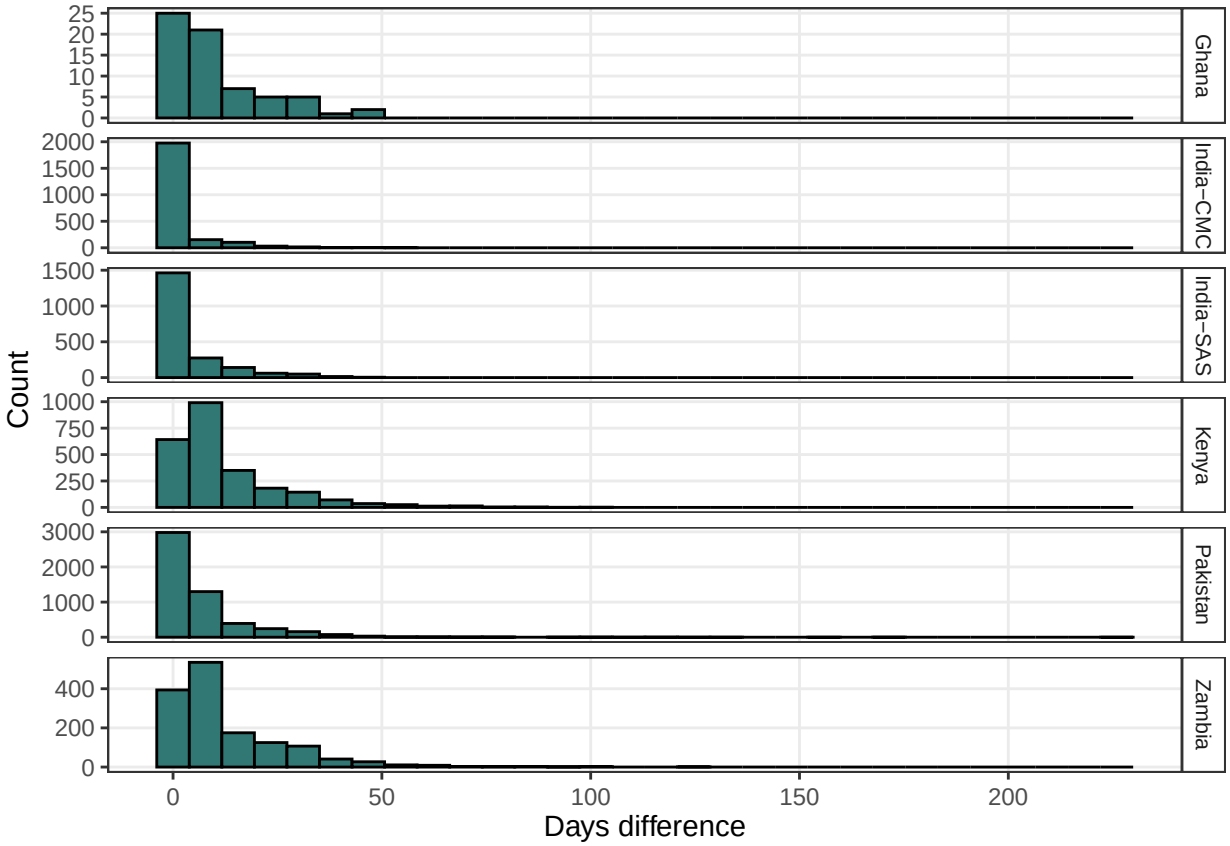


Figure 3b. Distribution of gestational age at birth in weeks.

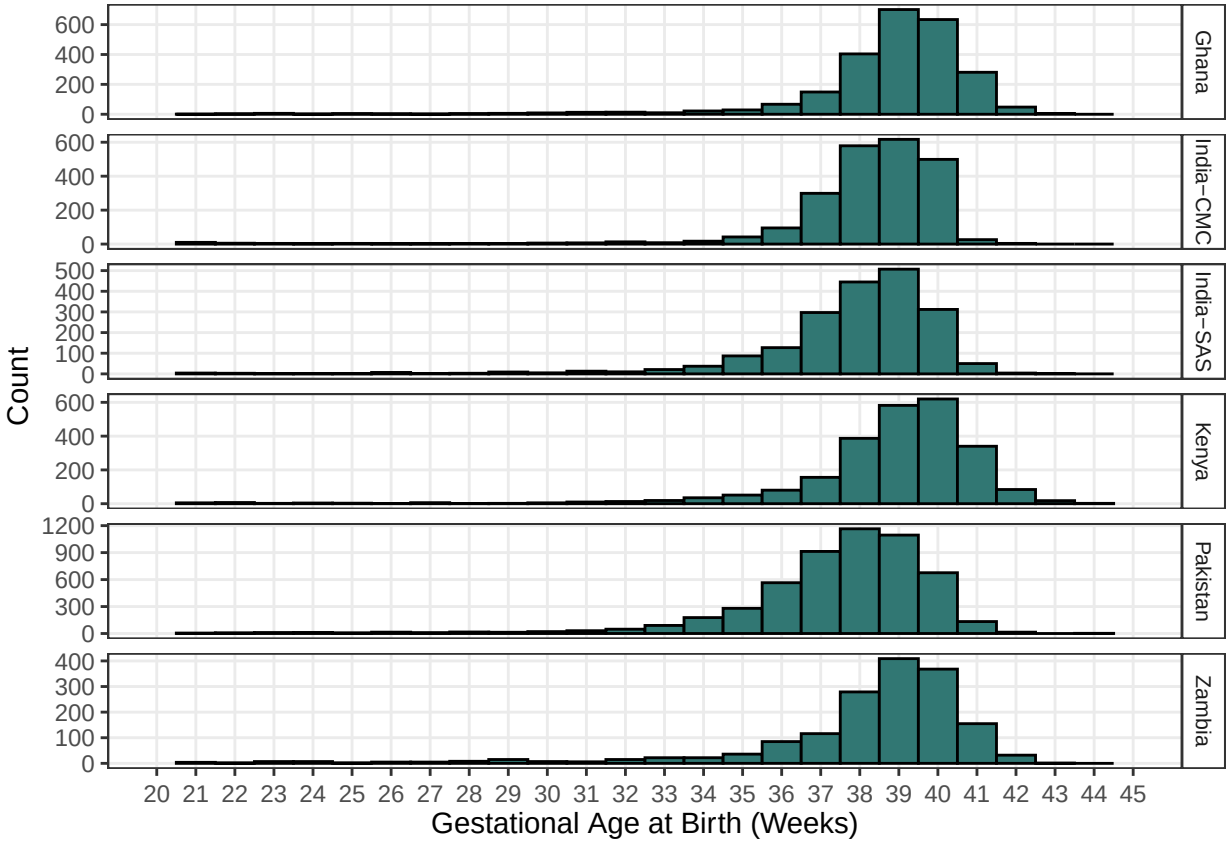


Figure 3b. Among all deliveries (stillbirths and livebirths). Gestational age at birth was calculated by taking the difference between the DOB and "estimated conception date" determined by BOE.

Figure 3c. Proportion of stillbirths and livebirths among all deliveries by gestational age where each week of gestation totals to 100% percent.

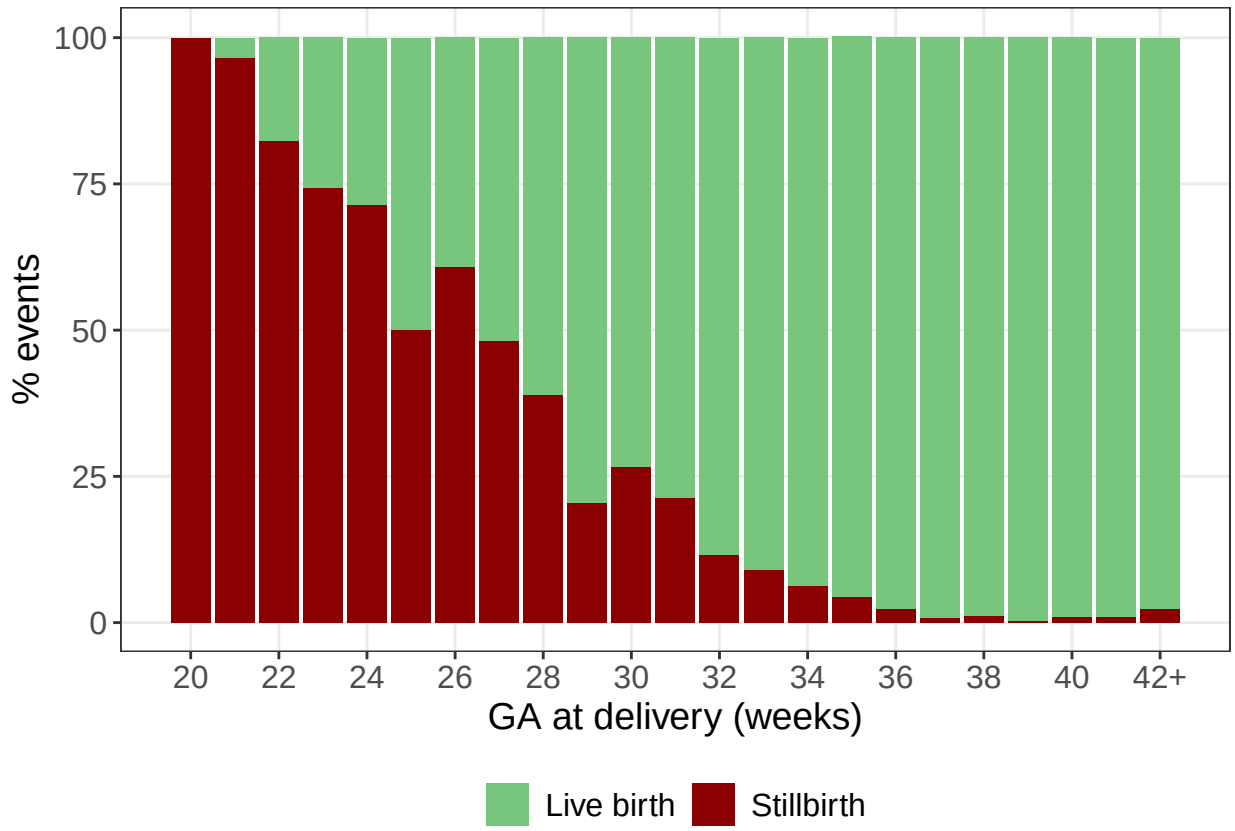
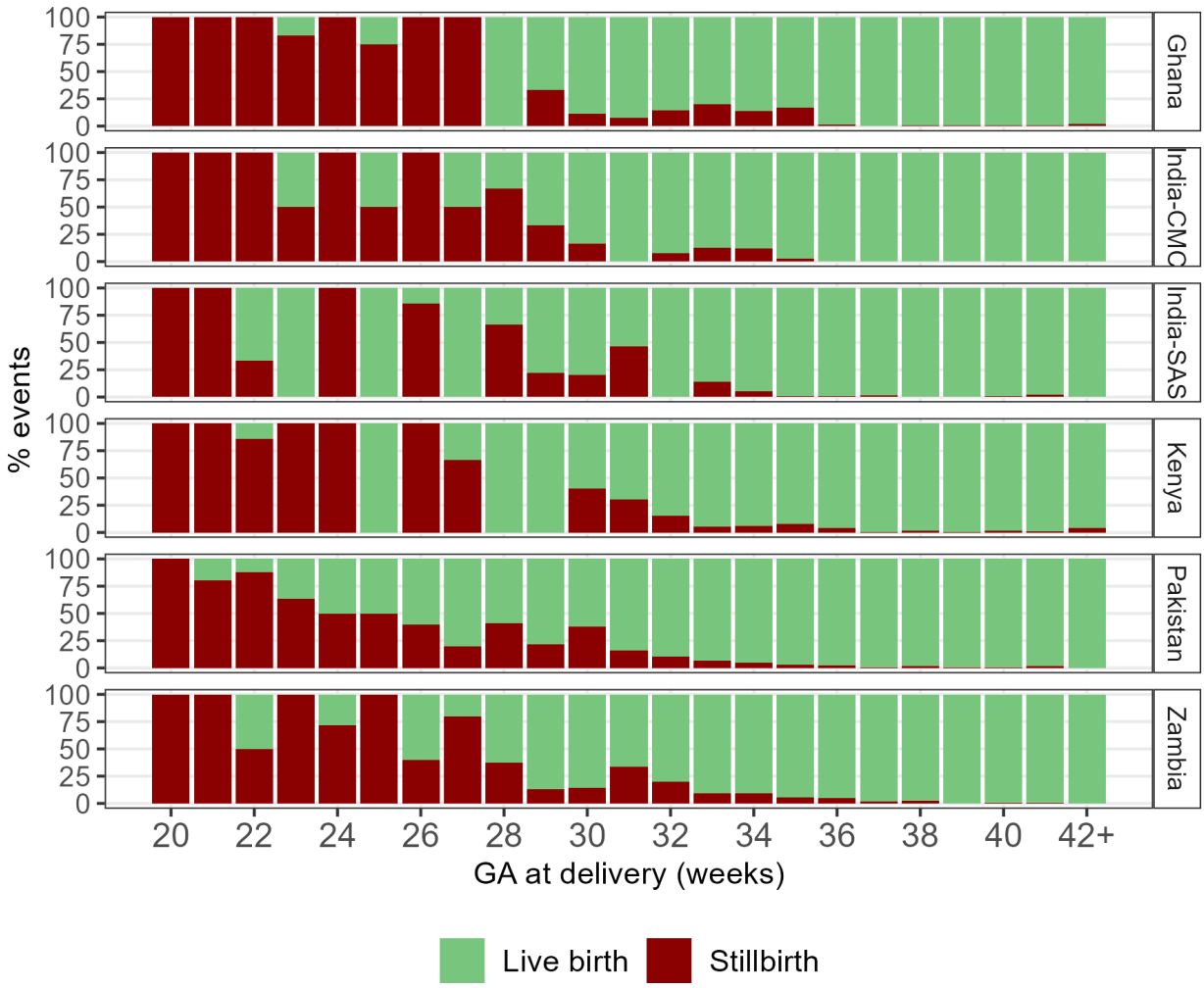


Figure 3d. By site, proportion of stillbirths and livebirths among all deliveries by gestational age where each week of gestation totals to 100% percent.



4. Size for gestational age (SGA/LGA)

Definition: GA & sex-specific birthweight percentiles per INTERGROWTH standard.Further classified as <3rd, 3rd to 10th, 10th to <90th, >=90th.

Denominator: All livebirths.

Note: *Gestational age information collected in MNH01 is used to generate best obstetric estimates for EDD. These constructed variables are then used to calculate GA at time of birth.*

To be included as "non-Missing" for this outcome, a participant must have:

1. Reported gestational age by either LMP or Ultrasound (varnames [form]: `US_GA_WKS_AGE_FTS1-4 [MNH01]`, `US_GA_DAYS_AGE_FTS1-4 [MNH01]`, `GA_LMP_WEEKS_SCORRES [MNH01]`).

2. Valid date of birth (varname: `DELIV_DSSTDAT_INF1-4 [MNH09]`).

3. Birthweight measured by PRISMA staff <72 hours following birth OR facility reported birthweight where PRISMA not available (varnames [form]: `BW_EST_FAORRES [MNH11]`, `BW_FAORRES [MNH11]`, `BW_FAORRES_REPORT [MNH11]`).

4. Live birth (varname: `BIRTH_DSTERM_INF1-4 [MNH09]`).

5. Sex of infant (varname: `SEX_INF1-4 [MNH09]`).

Common causes for a participant to be marked as "Missing":

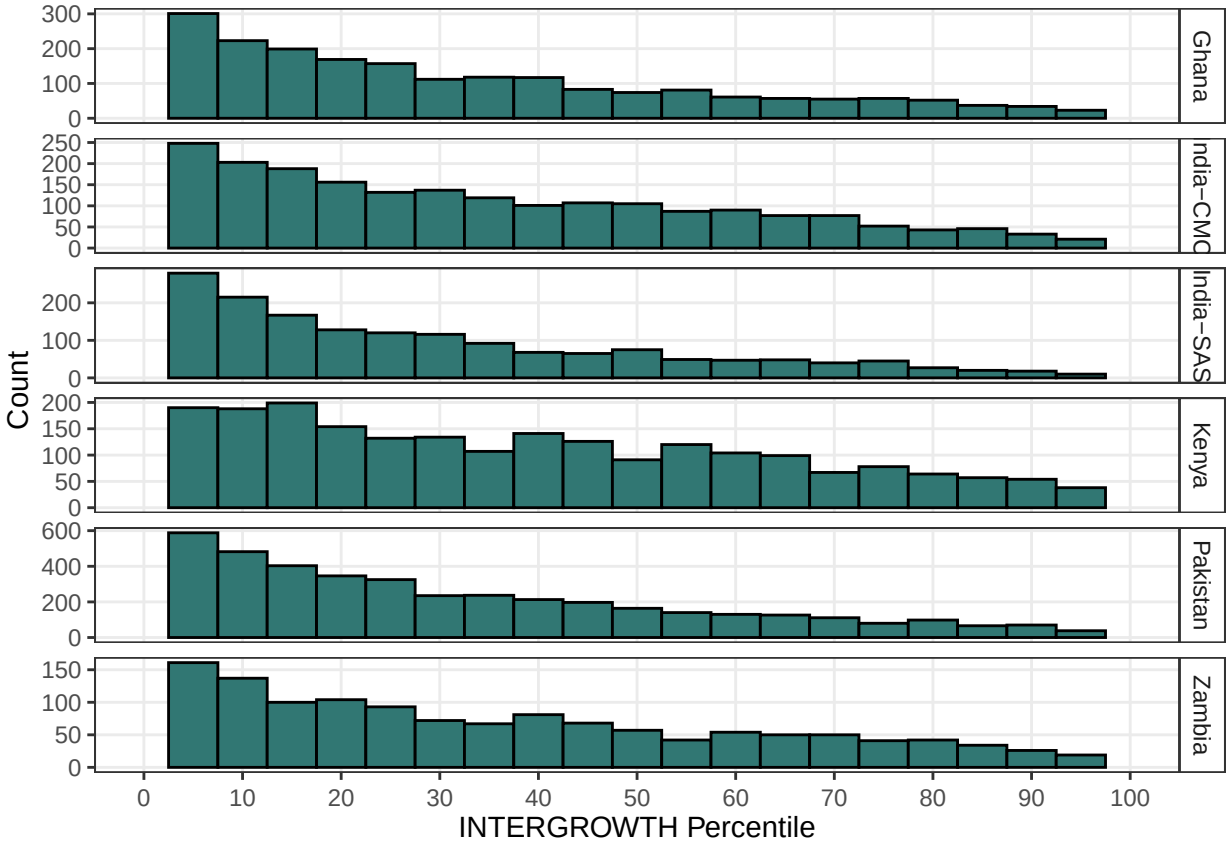
- Participant is missing a reported GA by Ultrasound AND GA by LMP in MNH01.
- Participant is missing an enrollment ultrasound visit (`TYPE_VISIT=1`).
- PRISMA-measured AND Facility-reported birthweights are missing from MNH11.
- MNH11 forms missing for infants (i.e. infantid present in MNH09, but is missing an MNH11 form).
- Gestational age at birth less than 33 weeks or over 42 weeks.
- INFANTID in MNH09 `(INFANTID_INF1-4)` that can be linked to MNH11 `(INFANTID)`.

Table 4. Size for gestational age at birth

Table 4						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Size for Gestational Age (Categorical)[a], n (%)						
Denominator	**2363**	**2212**	**1906**	**2372**	**5176**	**1552**
SGA <3rd percentile	262 (11)	174 (7.9)	261 (14)	172 (7.2)	580 (11)	152 (9.8)
SGA 3rd to <10th percentile	399 (17)	319 (14)	370 (19)	253 (11)	791 (15)	230 (15)
AGA 10th to <90th percentile	1570 (66)	1665 (75)	1241 (65)	1821 (77)	3186 (62)	1035 (67)
LGA >= 90th percentile	50 (2.1)	45 (2)	20 (1.0)	81 (3.4)	90 (1.7)	41 (2.6)

[a] Denominator is all livebirths.

Figure 4. Distribution of INTERGROWTH percentiles across sites.



5. Mortality

Definition: Death of a liveborn baby. Further stratified as neonatal mortality (death of a liveborn baby in the first 28 days of life) and infant mortality (death of a liveborn baby in the first 365 days of life).

Denominator: All livebirths AND have passed the risk period with a visit OR died within the risk period.

To be included as "non-Missing" for this outcome, a participant must have:

1. Valid date of birth (varname: `DELIV_DSSTDAT_INF1-4 [MNH09]`).

2. Birth outcome reported as "live birth" (varname: `BIRTH_DSTERM [MNH09]`)

Table 5. Mortality

Table 5						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Data Completeness[a]						
Total deaths reported <365 days	**38**	**17**	**51**	**52**	**327**	**65**
Death reported but missing date or time of death	2 (5.3)	0 (0)	0 (0)	13 (25)	10 (3.1)	12 (18)
Death reported but date or time of death is before delivery	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1 (1.5)
Timing of Neonatal Mortality[b]						
Denominator	**2363**	**2178**	**1906**	**2375**	**5091**	**1552**
Death <24 hrs of life	4 (0.17)	4 (0.18)	13 (0.68)	6 (0.25)	33 (0.65)	6 (0.39)
Death 24 hrs to <7 days of life	19 (0.8)	3 (0.14)	17 (0.89)	15 (0.63)	119 (2.3)	26 (1.7)
Death 7 days to <28 days of life	4 (0.17)	3 (0.14)	7 (0.37)	2 (0.08)	59 (1.2)	9 (0.58)
Total neonatal deaths	27 (1.1)	10 (0.46)	37 (1.9)	23 (0.97)	211 (4.1)	41 (2.6)
Timing of Post-neonatal Infant Mortality[c]						
Denominator	**1873**	**1259**	**1101**	**1787**	**3324**	**1463**
Death 28 days to <365 days of life	9 (0.48)	7 (0.56)	14 (1.3)	16 (0.9)	106 (3.2)	11 (0.75)
Total post-neonatal infant deaths	9 (0.48)	7 (0.56)	14 (1.3)	16 (0.9)	106 (3.2)	11 (0.75)

[a] Denominator is all reported deaths <365 days who have passed the risk period with a visit OR died within the risk period OR death was reported but date of death = missing OR date of death < DOB.

[b] Denominator is all livebirths with MNH09 and MNH11 filled out AND have passed the risk period with a visit OR died within the risk period (risk period: age = 28 days).

[c] Denominator is all livebirths with MNH09 and MNH11 filled out AND have passed the risk period (risk period: 365 days following birth).

6. Stillbirth

Definition: Death prior to delivery of a fetus at >=20 weeks of gestation, excluding induced abortions.

Denominator: All infants with a delivery outcome.

To be included as "non-Missing" for this outcome, a participant must have:

1. Valid date of birth OR valid fetal loss date (varname: `FETAL_LOSS_DSSTDAT [MNH04]` OR `DELIV_DSSTDAT_INF1-4 [MNH09]`).

2. Birth outcome reported in MNH04 or MNH11 (varname: `PRG_DSDECOD [MNH04]` OR `BIRTH_DSTERM [MNH09]`)

3. Reported gestational age by either LMP or Ultrasound (varnames [form]: `US_GA_WKS_AGE_FTS1-4 [MNH01]`, `US_GA_DAYS_AGE_FTS1-4 [MNH01]`, `GA_LMP_WEEKS_SCORRES [MNH01]`).

Table 6. Stillbirth

Table 6						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Data completeness[a] [b]						
Denominator	**2420**	**2257**	**1949**	**2444**	**5323**	**1611**
Fetal loss or fetal death reported but missing signs of life (NA or 77 reported)[a] [b]	3 (0.12)	1 (0.04)	0 (0)	61 (2.5)	0 (0)	0 (0)
Stillbirth (categorical)[c]						
Denominator	**2420**	**2257**	**1949**	**2444**	**5323**	**1611**
Early stillbirth (20-27wks)	21 (0.87)	34 (1.5)	14 (0.72)	30 (1.2)	48 (0.9)	28 (1.7)
Late stillbirth (28-36wks)	17 (0.7)	9 (0.4)	18 (0.92)	17 (0.7)	65 (1.2)	21 (1.3)
Term stillbirth (>=37wks)	19 (0.79)	2 (0.09)	11 (0.56)	22 (0.9)	34 (0.64)	10 (0.62)
Total stillbirths reported	57 (2.4)	45 (2)	43 (2.2)	69 (2.8)	147 (2.8)	59 (3.7)
Timing of stillbirth[d]						
Denominator	**57**	**45**	**43**	**69**	**147**	**59**
Antepartum	44 (77)	43 (96)	26 (60)	63 (91)	64 (44)	46 (78)
Intrapartum	2 (3.5)	0 (0)	17 (40)	3 (4.3)	8 (5.4)	3 (5.1)
Don't Know	11 (19)	2 (4.4)	0 (0)	3 (4.3)	75 (51)	10 (17)

[a] Denominator is all particpants with a birth outcome reported in MNH09.

[b] Valid signs of life reported in MNH09 and MNH11 (varname: `CRY_CEOCCUR_INF1-4 [MNH09]`, `FHR_VSTAT_INF1-4 [MNH09]`, `MACER_CEOCCUR_INF1-4 [MNH09]`, `CORD_PULS_CEOCCUR_INF1-4 [MNH09]`, `BREATH_FAIL_CEOCCUR [MNH11]`).

[c] Denominator is all participants with a birth outcome (`BIRTH_DSTERM=1 or 0 [MNH09]` OR `PRG_DSDECOD=2 or 3 [MNH04]`).

[d] Denominator is all reported stillbirths.

7. Fetal death

Definition: Death prior to delivery of a fetus at any gestational age.

Denominator: All infants with a delivery outcome.

To be included as "non-Missing" for this outcome, a participant must have:

1. Valid date of birth OR valid fetal loss date (varname: `FETAL_LOSS_DSSTDAT [MNH04]` OR `DELIV_DSSTDAT_INF1-4 [MNH09]`).

Table 7. Fetal death

Table 7	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Fetal death [a]						
Denominator	**2464**	**2282**	**1993**	**2478**	**5412**	**1625**
Stillbirth (fetal loss >=20wks)	57 (2.3)	45 (2)	43 (2.2)	69 (2.8)	147 (2.7)	59 (3.6)
Spontaneous abortion (fetal loss <20wks)	42 (1.7)	25 (1.1)	44 (2.2)	34 (1.4)	89 (1.6)	13 (0.8)
Fetal death at unknown GA	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Total fetal deaths	99 (4)	70 (3.1)	87 (4.4)	103 (4.2)	236 (4.4)	72 (4.4)
Other fetal death [b]						
Denominator	**2464**	**2290**	**2008**	**2478**	**5416**	**1625**
Induced abortion [c]	0 (0)	8 (0.35)	15 (0.75)	0 (0)	4 (0.07)	0 (0)

[a] Denominator is n total deliveries excluding induced abortions

[b] Denominator is n total deliveries.

[c] Induced abortion defined as an elective surgical procedure or medical intervention to terminate the pregnancy at any gestational age).

8. Perinatal birth asphyxia

Definition: Clinician reports failure to breathe spontaneously in the first minute after delivery or breathing assistance was required.

Denominator: All livebirths.

To be included as "non-Missing" for this outcome, a participant must have:

1. Valid birth complications reported in MNH11 or MNH20 (varname: `INF_PROCCUR_1-6 [MNH11]`)

2. Valid birth complications reported in MNH20, if applicable (varname: `BIRTH_COMPL_MHTERM_3 [MNH20]`)

Table 8. Perinatal birth asphyxia

Table 8						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Data completeness[a]						
Denominator	**2363**	**2212**	**1906**	**2372**	**5176**	**1552**
Invalid birth complications in MNH11 (NA or 77 reported)[a]	16 (0.68)	2 (0.09)	4 (0.21)	8 (0.34)	0 (0)	0 (0)
Indicators of birth asphyxia[a]						
Denominator	**2363**	**2212**	**1906**	**2372**	**5176**	**1552**
Failed to initiate and sustain breathing at birth	113 (4.8)	107 (4.8)	202 (11)	111 (4.7)	530 (10)	18 (1.2)
Bag and mask ventilation	61 (2.6)	52 (2.4)	14 (0.73)	29 (1.2)	29 (0.56)	8 (0.52)
Continuous positive airway pressure	20 (0.85)	28 (1.3)	23 (1.2)	17 (0.72)	25 (0.48)	62 (4)
Repeated stimulation/suction	138 (5.8)	32 (1.4)	11 (0.58)	30 (1.3)	2 (0.04)	54 (3.5)
Intubation and mechanical ventilation	25 (1.1)	11 (0.5)	12 (0.63)	0 (0)	51 (0.99)	3 (0.19)
Chest compressions	44 (1.9)	2 (0.09)	1 (0.05)	9 (0.38)	4 (0.08)	3 (0.19)
Perinatal birth asphyxia[a]						
Denominator	**2363**	**2212**	**1906**	**2372**	**5176**	**1552**
Birth asphyxia[b]	255 (11)	154 (7)	216 (11)	115 (4.8)	535 (10)	128 (8.2)

[a] Denominator is total liveborn infants with a MNH11 form filled out.

[b] Defined as required any breathing assistance at birth.

9. Hyperbilirubinemia

Definition: Defined as the presence of excess bilirubin during the first week of life (delivery to 7 days of age).

Denominator: All liveborn infants with a measurement taken within the testing window.

Table 9. Hyperbilirubinemia

Table 9a (continued on next page)

	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Hyperbilirubinemia by TCB (>15mg/dL)						
Hyperbilirubinemia <24 hrs of life	4 (0.32)	1 (0.11)	5 (0.76)	8 (0.91)	0 (0)	0 (0)
Hyperbilirubinemia 24 hrs to <5 days of life	137 (8.5)	204 (11)	170 (11)	164 (7.7)	408 (9)	62 (6.5)
Hyperbilirubinemia 5 days to <14 days of life	68 (3.9)	21 (1.2)	126 (7)	72 (3.4)	283 (6.4)	32 (4.6)
Total hyperbilirubinemia by TCB >15mg/dL	209 (9.3)	226 (11)	301 (16)	244 (10)	691 (14)	94 (6.5)
Hyperbilirubinemia by TCB (AAP threshold) [a]						
Hyperbilirubinemia <24 hrs of life	1 (0.08)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Hyperbilirubinemia 24 hrs to <5 days of life	33 (2.1)	82 (4.3)	49 (3.1)	33 (1.5)	133 (2.9)	24 (2.5)
Hyperbilirubinemia 5 days to <14 days of life	17 (0.97)	10 (0.57)	38 (2.1)	25 (1.2)	97 (2.2)	7 (1)
Total hyperbilirubinemia by AAP threshold	51 (2.3)	92 (4.3)	87 (4.5)	58 (2.4)	230 (4.5)	31 (2.1)

[a] AAP and NICE threshold determined by the following inputs: phototherapy thresholds with no hyperbilirubinemia neurotoxicity risk factor, gestational age at delivery, and infant age at time of assessment (days and hours).

Table 9a continued						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Hyperbilirubinemia by TCB (NICE threshold) [a]						
Hyperbilirubinemia <24 hrs of life	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Hyperbilirubinemia 24 hrs to <5 days of life	47 (2.9)	189 (9.8)	138 (8.7)	93 (4.3)	442 (9.7)	33 (3.5)
Hyperbilirubinemia 5 days to <14 days of life	38 (2.2)	35 (2)	138 (7.6)	59 (2.8)	342 (7.7)	23 (3.3)
Total hyperbilirubinemia by NICE threshold	85 (3.8)	224 (10)	276 (14)	152 (6.3)	784 (15)	56 (3.9)
Jaundice (IMCI criteria)						
Non-severe jaundice identified at any time	116 (5.1)	277 (13)	220 (12)	257 (11)	1838 (37)	104 (7.1)
Severe jaundice identified <24 hrs of life	6 (0.27)	3 (0.14)	4 (0.21)	13 (0.56)	3 (0.06)	1 (0.07)
Severe jaundice identified >=24 hrs of life	58 (2.6)	164 (7.9)	101 (5.4)	62 (2.7)	498 (10)	58 (3.9)
Any jaundice identified at any time	171 (7.6)	428 (21)	299 (16)	311 (13)	2149 (43)	155 (11)

[a] AAP and NICE threshold determined by the following inputs: phototherapy thresholds with no hyperbilirubinemia neurotoxicity risk factor, gestational age at delivery, and infant age at time of assessment (days and hours).

10. PSBI

Definition: Presence of any clinical signs or symptoms as defined by the WHO IMCI criteria, which are consistent with possible severe bacterial infection from delivery to 59 days.

Denominator: All livebirths aged 0-59 days.

Table 10. PSBI

Table 10						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Data Completeness: missing confirmatory respiratory rate measurement [a]						
IPC	20 (12)	1 (0.51)	5 (2.7)	14 (9.8)	1 (0.17)	2 (3.9)
PNC0	41 (34)	1 (0.75)	0 (0)	0 (0)	0 (0)	8 (14)
PNC1	54 (46)	2 (2.7)	0 (0)	0 (0)	0 (0)	13 (31)
PNC4	29 (40)	4 (13)	0 (0)	1 (5.6)	0 (0)	4 (36)
PNC6	9 (75)	0 (0)	0 (0)	1 (4)	0 (0)	5 (17)
Unscheduled visit	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	3 (27)
Hospitalization visit	0 (0)	44 (23)	28 (58)	4 (6.4)	7 (12)	0 (0)
Data Completeness: missing confirmatory temperature measurement [b]						
IPC	11 (6.4)	2 (1)	0 (0)	12 (8.4)	0 (0)	6 (12)
PNC0	25 (21)	1 (0.75)	0 (0)	11 (12)	0 (0)	24 (42)
PNC1	22 (19)	0 (0)	0 (0)	4 (11)	0 (0)	17 (40)
PNC4	18 (25)	0 (0)	0 (0)	6 (33)	0 (0)	3 (27)
PNC6	0 (0)	0 (0)	0 (0)	5 (20)	0 (0)	12 (41)
Unscheduled visit	0 (0)	0 (0)	0 (0)	1 (20)	0 (0)	1 (9.1)
Hospitalization visit	0 (0)	168 (6.3)	30 (12)	40 (31)	30 (29)	34 (15)

[a] Missing confirmatory respiratory rate measurements among those with a RR>=60 breaths per minute. Denominator is n total psbi cases at the specified visit.

[b] Missing confirmatory temperature among those with a temperature reading >=38C or <35.5C. Denominator is n total psbi cases at the specified visit.

Table 10 continued						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
PSBI by visit [c]						
Denominator	2363	2212	1906	2375	5176	1552
IPC (target window: delivery to 72 hours)	173 (7.3)	198 (8.9)	185 (9.7)	143 (6)	598 (12)	51 (3.3)
Difficulty in feeding	14	34	14	49	48	17
Convulsions	5	3	4	2	27	10
Fast breathing (RR>=60)	61	127	18	28	325	3
Severe chest in-drawing	22	20	11	30	12	23
Fever (>=38C)	21	2	9	61	107	2
Low body temperature (<35.5C)	67	37	138	6	142	5
PNC0 (target window: 3 to 5 days)	119 (5)	133 (6)	417 (22)	88 (3.7)	287 (5.5)	57 (3.7)
Difficulty in feeding	6	17	1	19	2	7
Convulsions	4	2	4	2	2	1
Fast breathing (RR>=60)	59	73	9	7	144	17
Severe chest in-drawing	8	8	3	11	0	5
Fever (>=38C)	42	5	25	62	91	22
Low body temperature (<35.5C)	6	40	380	5	75	12
PNC1 (target window: 7 to 14 days)	118 (5)	73 (3.3)	414 (22)	35 (1.5)	261 (5)	42 (2.7)
Difficulty in feeding	0	13	2	12	2	3
Convulsions	3	3	1	1	1	0
Fast breathing (RR>=60)	70	40	7	5	86	17
Severe chest in-drawing	4	1	2	5	0	2
Fever (>=38C)	38	5	31	22	57	16
Low body temperature	6	20	381	0	122	9

Table 10 continued	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
PSBI by visit [c]						
PNC4 (target window: 28 to 35 days)	73 (3.1)	30 (1.4)	445 (23)	18 (0.76)	194 (3.8)	11 (0.71)
Difficulty in feeding	3	8	2	1	0	0
Convulsions	5	1	0	0	0	0
Fast breathing (RR>=60)	41	13	7	3	66	8
Severe chest in-drawing	3	5	8	1	3	0
Fever (>=38C)	19	2	27	13	32	1
Low body temperature (<35.5C)	5	6	408	3	99	3
PNC6 (target window: 6 to 12 weeks)	12 (0.51)	35 (1.6)	430 (23)	25 (1.0)	132 (2.5)	29 (1.9)
Difficulty in feeding	1	3	3	4	0	1
Convulsions	0	1	1	0	0	0
Fast breathing (RR>=60)	9	4	8	1	72	9
Severe chest in-drawing	2	4	4	0	5	0
Fever (>=38C)	1	10	19	17	29	8
Low body temperature (<35.5C)	0	14	403	3	31	11

[c] Denominator is all livebirths aged 0-59 days.

Table 10 continued						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
PSBI by visit [c]						
Unscheduled	0 (0)	0 (0)	0 (0)	5 (0.21)	10 (0.19)	11 (0.71)
Difficulty in feeding	0	0	0	2	0	1
Convulsions	0	0	0	0	0	0
Fast breathing (RR>=60)	0	0	0	0	4	6
Severe chest in-drawing	0	0	0	0	0	3
Fever (>=38C)	0	0	0	3	1	2
Low body temperature (<35.5C)	0	0	0	0	5	1
Hospitalization	0 (0)	191 (8.6)	48 (2.5)	62 (2.6)	58 (1.1)	78 (5)
Difficulty in feeding	0	23	21	40	17	47
Convulsions	0	10	14	6	11	12
Fast breathing (RR>=60)	0	155	30	11	8	13
Severe chest in-drawing	0	24	1	12	16	22
Fever (>=38C)	0	27	5	21	19	16
Low body temperature (<35.5C)	0	6	1	2	0	4

[c] Denominator is all livebirths aged 0-59 days.

11. Neonatal Sepsis

Definition: Inflammatory response and organ dysfunction following presence of a severe infection from delivery to 28 days as suspected (by a clinician) or proven (with culture). Defined below using reported neonatal sepsis variables and confirmed blood culture (varnames [form]: `INFANT_MHTERM_10 [MNH11]`, `INFANT_MHTERM_10 [MNH13]`, `INFECTION_MHTERM_1 [MNH20]`,`CULTURE_LBORRES [MNH11]`, `BLD_CULT_LBORRES [MNH13]`, `BLD_CULT_LBORRES [MNH20]`).

Denominator: All livebirths.

Table 11. Neonatal Sepsis

Table 11	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Neonatal Sepsis						
Denominator	**2363**	**2212**	**1906**	**2372**	**5176**	**1552**
Neonatal sepsis[a]	10 (0.42)	21 (0.95)	6 (0.31)	37 (1.6)	26 (0.5)	95 (6.1)
Confirmed positive blood culture[b]	0 (0)	6 (29)	0 (0)	7 (19)	1 (3.8)	0 (0)
Concurrent PSBI[b c]	1 (10)	2 (9.5)	0 (0)	15 (41)	13 (50)	26 (27)

[a] Denominator is all livebirths.

[b] Denominator is all reported neonatal sepsis cases.

[c] PSBI defined as the presence of any clinical signs or symptoms as defined by the WHO IMCI criteria, which are consistent with possible severe bacterial infection from delivery to 59 days (Table 10, above).

12. Infant growth trajectories

Definitions:

- Underweight (Weight for Age - WAZ): Proportion of infants more than 2 standard deviations (z score) below the WHO Children Health Reference Growth Standard.

- Severely Underweight (Weight for Age - WAZ): Proportion of infants more than 3 standard deviations (z score) below the WHO Children Health Reference Growth Standard.

- Wasting (Weight for Length - WFL): Proportion of infants with more than 2 standard deviations (z score) below the WHO Children Health Reference Growth Standard.

- Severe Wasting (Weight for Length - WFL): Proportion of infants more than 3 standard deviations (z score) below the WHO Children Health Reference Growth Standard.

- Stunting (Length for Age - LAZ): Proportion of infants more than 2 standard deviations (z score) below the WHO Children Health Reference Growth Standard

- Severe Stunting (Length for Age - LAZ): Proportion of infants more than 3 standard deviations (z score) below the WHO Children Health Reference Growth Standard

- Microcephaly (Head circumference for Age - HCZ): Proportion of infants more than 3 standard deviations (z score) below the WHO Children Health Reference Growth Standard.

- Overweight (Weight for Length - WFL): Proportion of infants with more than 2 standard deviations (z score) below the WHO Children Health Reference Growth Standard.

Table 12a. Infant growth outcome at birth

Denominator: All liveborn infants who have completed the IPC visit (0-5 days) (MNH11 & MNH09) AND have all anthropometric measures AND have valid z-scores.

Table 12a						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Data completeness at birth (0 – 5 days)						
Denominator [a]	**2363**	**2212**	**1906**	**2372**	**5176**	**1552**
Birthdate–Age Inconsistencies [b]	182 (7.7)	226 (10)	88 (4.6)	37 (1.6)	409 (7.9)	106 (6.8)
Missing information on infant sex	0 (0)	0 (0)	1 (0.05)	0 (0)	1 (0.02)	5 (0.32)
Missing information on infant length	85 (3.6)	327 (15)	328 (17)	158 (6.7)	440 (8.5)	239 (15)
Missing information on infant weight	72 (3)	7 (0.32)	6 (0.31)	25 (1.0)	523 (10)	87 (5.6)
Missing information on head circumference	84 (3.5)	336 (15)	325 (17)	142 (6)	421 (8.1)	237 (15)
Missing at least one anthropometry measurement	236 (10)	354 (16)	333 (17)	164 (6.9)	962 (19)	251 (16)
Invalid Z score [c d]	152 (6.4)	82 (3.7)	65 (3.4)	260 (11)	471 (9.1)	128 (8.2)

[a] Denominator is total livebirths with MNH09 & MNH11.

[b] Birthdate–Age Inconsistencies: includes records where the timing of birth anthropometric measurement implies (i) a negative calculated age, (ii) age outside the allowable birth visit window (>5 days), or (iii) missing or implausible visit dates (e.g., before 2021 or in the future). These are generally a result of conflicts between site-provided MNH11 visit date and the MNH09 recorded birthdate.

[c] Participants who had at least one invalid z-score measure.

[d] Invalid Z-score - At least one of the z-scores has been flagged by the World Health Organization (WHO) as biologically implausible (BIV).

Table 12a continued						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Infant growth outcomes at birth (0 – 5 days) using WHO Standards						
Denominator [e]	**1975**	**1776**	**1508**	**1948**	**3743**	**1173**
Underweight (WAZ < -2)	155 (7.8)	138 (7.8)	233 (15)	78 (4)	464 (12)	88 (7.5)
Severe underweight (WAZ <-3)	26 (1.3)	11 (0.62)	31 (2.1)	6 (0.31)	32 (0.85)	15 (1.3)
Wasting (WFL < -2)	278 (14)	146 (8.2)	218 (14)	109 (5.6)	309 (8.3)	209 (18)
Severe wasting (WFL < -3)	101 (5.1)	21 (1.2)	67 (4.4)	30 (1.5)	65 (1.7)	78 (6.6)
Stunting (LAZ <- 2)	135 (6.8)	144 (8.1)	113 (7.5)	167 (8.6)	450 (12)	54 (4.6)
Severe Stunting (LAZ <- 3)	0 (0)	2 (0.11)	0 (0)	0 (0)	0 (0)	0 (0)
Overweight (WFL > 3)	66 (3.3)	14 (0.79)	25 (1.7)	136 (7)	28 (0.75)	42 (3.6)
Microcephaly (HCZ < -3)	46 (2.3)	49 (2.8)	24 (1.6)	2 (0.1)	65 (1.7)	6 (0.51)
Infant growth outcomes at birth (0 – 5 days) using Intergrowth Standard (adjusting for preterm deliveries)						
Underweight (WAZ < -2)	151 (7.6)	81 (4.6)	159 (11)	80 (4.1)	272 (7.3)	76 (6.5)
Severe underweight (WAZ <-3)	10 (0.51)	2 (0.11)	13 (0.86)	4 (0.21)	14 (0.37)	2 (0.17)
Wasting (WFL < -2)	178 (9)	97 (5.5)	166 (11)	86 (4.4)	269 (7.2)	101 (8.6)
Severe wasting (WFL < -3)	30 (1.5)	7 (0.39)	23 (1.5)	7 (0.36)	22 (0.59)	8 (0.68)
Stunting (LAZ <- 2)	94 (4.8)	40 (2.2)	43 (2.8)	145 (7.4)	116 (3.1)	38 (3.2)
Severe Stunting (LAZ <- 3)	1 (0.05)	0 (0)	1 (0.07)	5 (0.26)	1 (0.03)	0 (0)
Overweight (WFL > 3)	21 (1.1)	5 (0.28)	5 (0.33)	24 (1.2)	25 (0.67)	14 (1.2)
Microcephaly (HCZ < -3)	20 (1)	18 (1)	6 (0.4)	2 (0.1)	1 (0.03)	4 (0.34)

[d] All living infants who have completed the IPC visit (0-5 days) (have MNH11 & MNH09) AND have all anthropometric measures AND have valid z-scores.

Table 12b. Infant growth outcome at one month

Denominator: All liveborn infants who have completed at least one visit at one month (21-42 days) of age, AND have all required anthropometric measurements, AND possess valid z-scores.

Table 12b	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Data completeness at one month (21 to 42 days)						
Denominator [a]	**2052**	**2014**	**1863**	**2157**	**4289**	**1444**
Missing information on infant length	7 (0.34)	337 (17)	134 (7.2)	8 (0.37)	13 (0.3)	157 (11)
Missing information on infant weight	6 (0.29)	337 (17)	134 (7.2)	5 (0.23)	43 (1)	444 (31)
Missing information on head circumference	5 (0.24)	337 (17)	134 (7.2)	7 (0.32)	2 (0.05)	157 (11)
Missing at least one anthropometry measurement	8 (0.39)	337 (17)	134 (7.2)	8 (0.37)	52 (1.2)	444 (31)
Invalid Z score [b] [c]	31 (1.5)	28 (1.4)	20 (1.1)	31 (1.4)	120 (2.8)	24 (1.7)

[a] Denominator is any infant who have completed any visit within 21 to 42 days.

[b] Participants who had at least one invalid z-score measure.

[c] Invalid Z-score - At least one of the z-scores has been flagged by the World Health Organization (WHO) as biologically implausible (BIV).

Table 12b continued						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Infant growth outcomes in the first month of life using WHO Growth Standards						
Denominator [d]	**2013**	**1649**	**1709**	**2118**	**4117**	**976**
Underweight (WAZ < -2)	123 (6.1)	242 (15)	280 (16)	87 (4.1)	773 (19)	65 (6.7)
Severe underweight (WAZ <-3)	32 (1.6)	55 (3.3)	70 (4.1)	22 (1)	222 (5.4)	16 (1.6)
Wasting (WFL < -2)	235 (12)	155 (9.4)	204 (12)	89 (4.2)	538 (13)	95 (9.7)
Severe wasting (WFL < -3)	76 (3.8)	29 (1.8)	43 (2.5)	27 (1.3)	162 (3.9)	25 (2.6)
Stunting (LAZ <- 2)	78 (3.9)	179 (11)	199 (12)	126 (6)	671 (16)	122 (12)
Severe Stunting (LAZ <- 3)	9 (0.45)	32 (1.9)	31 (1.8)	22 (1)	142 (3.4)	25 (2.6)
Overweight (WFL > 3)	25 (1.2)	4 (0.24)	17 (0.99)	131 (6.2)	65 (1.6)	79 (8.1)
Microcephaly (HCZ < -3)	9 (0.45)	13 (0.79)	23 (1.4)	3 (0.14)	109 (2.6)	6 (0.61)
Infant growth outcomes in the first month of life using Intergrowth Standard (adjusting for preterm deliveries)						
Underweight (WAZ < -2)	123 (6.1)	242 (15)	280 (16)	87 (4.1)	773 (19)	65 (6.7)
Severe underweight (WAZ <-3)	32 (1.6)	55 (3.3)	70 (4.1)	22 (1)	222 (5.4)	16 (1.6)
Wasting (WFL < -2)	235 (12)	155 (9.4)	204 (12)	89 (4.2)	538 (13)	95 (9.7)
Severe wasting (WFL < -3)	76 (3.8)	29 (1.8)	43 (2.5)	27 (1.3)	162 (3.9)	25 (2.6)
Stunting (LAZ <- 2)	78 (3.9)	179 (11)	199 (12)	126 (6)	671 (16)	122 (12)
Severe Stunting (LAZ <- 3)	9 (0.45)	32 (1.9)	31 (1.8)	22 (1)	142 (3.4)	25 (2.6)
Overweight (WFL > 3)	25 (1.2)	4 (0.24)	17 (0.99)	131 (6.2)	65 (1.6)	79 (8.1)
Microcephaly (HCZ < -3)	9 (0.45)	13 (0.79)	23 (1.4)	3 (0.14)	109 (2.6)	6 (0.61)

[d] All living infants who have completed at least one visit at one month (21-42 days) of age, AND have all required anthropometric measurements, AND possess valid z-scores.

Table 12c. Infant growth outcome at six months

Denominator: All liveborn infants who have completed at least one visit at six months of age (171-222 days) AND have all required anthropometric measurements AND possess valid z-scores.

Table 12c						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Data completeness at six months (171 to 222 days)						
Denominator [a]	**1192**	**1331**	**1799**	**1818**	**3500**	**1366**
Missing information on infant length	2 (0.17)	125 (9.4)	132 (7.3)	8 (0.44)	5 (0.14)	146 (11)
Missing information on infant weight	2 (0.17)	125 (9.4)	132 (7.3)	1 (0.06)	44 (1.3)	541 (40)
Missing information on head circumference	3 (0.25)	125 (9.4)	131 (7.3)	8 (0.44)	1 (0.03)	146 (11)
Missing at least one anthropometry measurement	3 (0.25)	125 (9.4)	132 (7.3)	8 (0.44)	48 (1.4)	541 (40)
Invalid Z score [b] [c]	22 (1.8)	9 (0.68)	20 (1.1)	9 (0.5)	106 (3)	21 (1.5)

[a] Denominator is any infant who have completed any visit within 171 to 222 days.

[b] Participants who had at least one invalid z-score measure.

[c] Invalid Z-score - At least one of the z-scores has been flagged by the World Health Organization (WHO) as biologically implausible (BIV).

Table 12c continued						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Infant growth outcomes at six months using WHO Growth Standards						
Denominator [d]	**1167**	**1197**	**1647**	**1801**	**3346**	**804**
Underweight (WAZ < -2)	127 (11)	147 (12)	275 (17)	128 (7.1)	1078 (32)	89 (11)
Severe underweight (WAZ <-3)	28 (2.4)	23 (1.9)	67 (4.1)	40 (2.2)	362 (11)	20 (2.5)
Wasting (WFL < -2)	115 (9.8)	73 (6.1)	166 (10)	92 (5.1)	696 (21)	89 (11)
Severe wasting (WFL < -3)	19 (1.6)	8 (0.67)	31 (1.9)	22 (1.2)	229 (6.8)	33 (4.1)
Stunting (LAZ <- 2)	90 (7.7)	125 (10)	206 (13)	141 (7.8)	729 (22)	169 (21)
Severe Stunting (LAZ <- 3)	23 (2)	20 (1.7)	43 (2.6)	31 (1.7)	230 (6.9)	50 (6.2)
Overweight (WFL > 3)	19 (1.6)	13 (1.1)	55 (3.3)	116 (6.4)	30 (0.9)	67 (8.3)
Microcephaly (HCZ < -3)	14 (1.2)	23 (1.9)	56 (3.4)	8 (0.44)	204 (6.1)	13 (1.6)
Infant growth outcomes at six months using Intergrowth Standard (adjusting for preterm deliveries)						
Underweight (WAZ < -2)	116 (9.9)	131 (11)	239 (15)	110 (6.1)	941 (28)	82 (10)
Severe underweight (WAZ <-3)	25 (2.1)	17 (1.4)	50 (3)	26 (1.4)	303 (9.1)	16 (2)
Wasting (WFL < -2)	115 (9.8)	73 (6.1)	166 (10)	92 (5.1)	696 (21)	89 (11)
Severe wasting (WFL < -3)	19 (1.6)	8 (0.67)	31 (1.9)	22 (1.2)	229 (6.8)	33 (4.1)
Stunting (LAZ <- 2)	82 (7)	102 (8.5)	164 (10)	117 (6.5)	598 (18)	150 (19)
Severe Stunting (LAZ <- 3)	18 (1.5)	15 (1.2)	25 (1.5)	21 (1.2)	174 (5.2)	40 (5)
Overweight (WFL > 3)	19 (1.6)	13 (1.1)	55 (3.3)	116 (6.4)	30 (0.9)	67 (8.3)
Microcephaly (HCZ < -3)	8 (0.69)	19 (1.6)	42 (2.5)	7 (0.39)	186 (5.6)	8 (1)

[d] All living infants who have completed at least one visit at six months of age (171-222 days) AND have all required anthropometric measurements AND possess valid z-scores.

Infant growth visualizations

The following distributions visualizes the mean Length for Age (LAZ), Weight for Age (WAZ) and Head Circumference (HC) Z-scores by site

Z-scores were generated using the

1. Intergrowth Standard
2. WHO Standard

For IPC Visits, the Intergrowth package calculates Z-scores for all infants based on the Gestational Age at birth. However, for subsequent visits, the Intergrowth package ONLY ADJUSTS FOR PRETERM babies. This adjustment is only available for;

- Weight for Age
- Length for Age
- Head Circumference for Age

There is no r-package available to adjust for preterm Weight-for-Length z-score after birth

PRISMA Visit Windows:

Birth: Initial measurement at birth;

PNC-4: Postnatal check at 28-35 days;

PNC-6: Postnatal check at 6-12 weeks;

PNC-26: Postnatal check at 26-39 weeks;

PNC-52: Postnatal check at 52-64 weeks.

Abbreviations:

LAZ: Length-for-Age Z-score, WAZ: Weight-for-Age Z-score,

WLF: Weight-for-length, HC: Head Circumference

Figure 5: Mean infant growth measures by PRISMA PNC visit windows across sites.

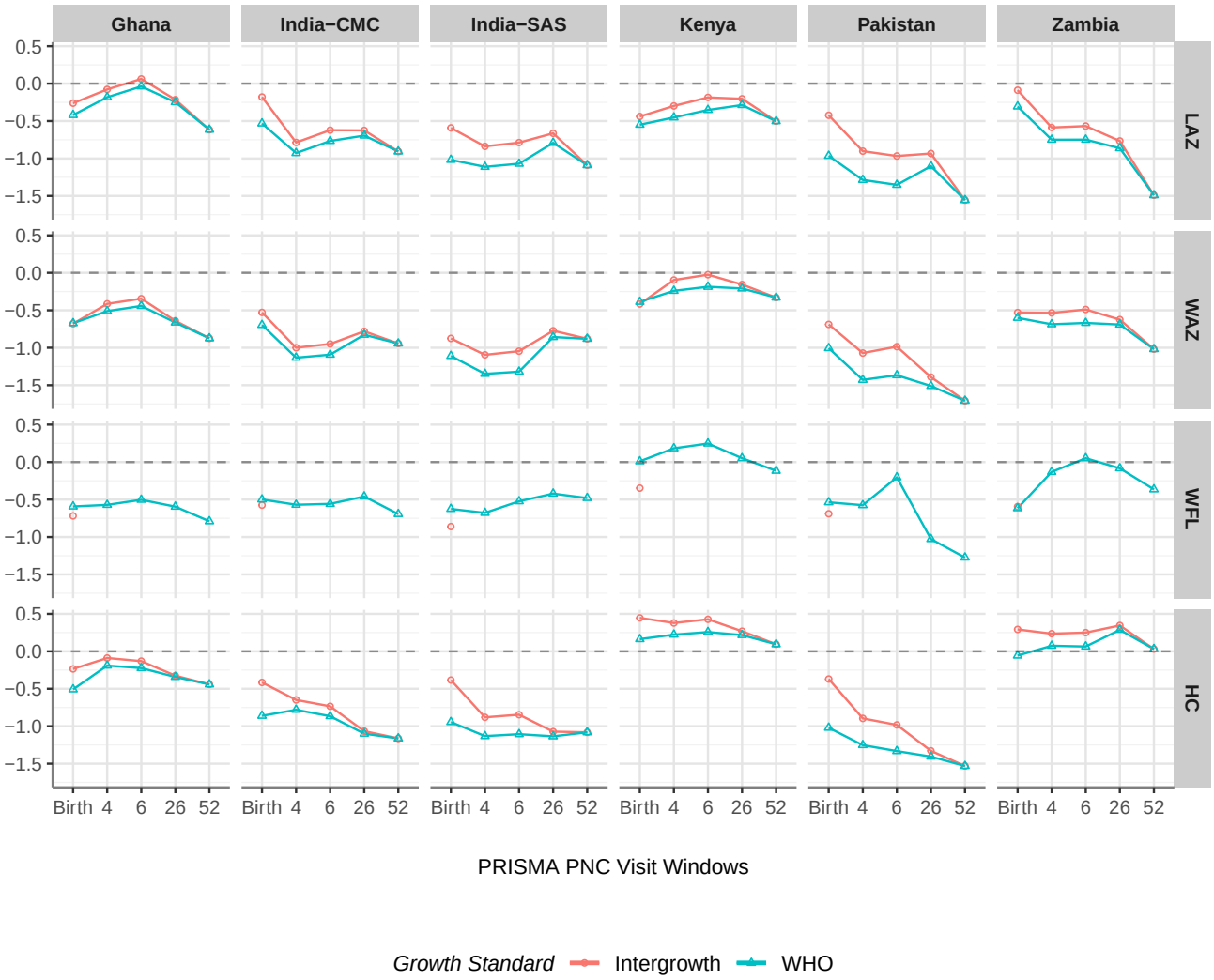
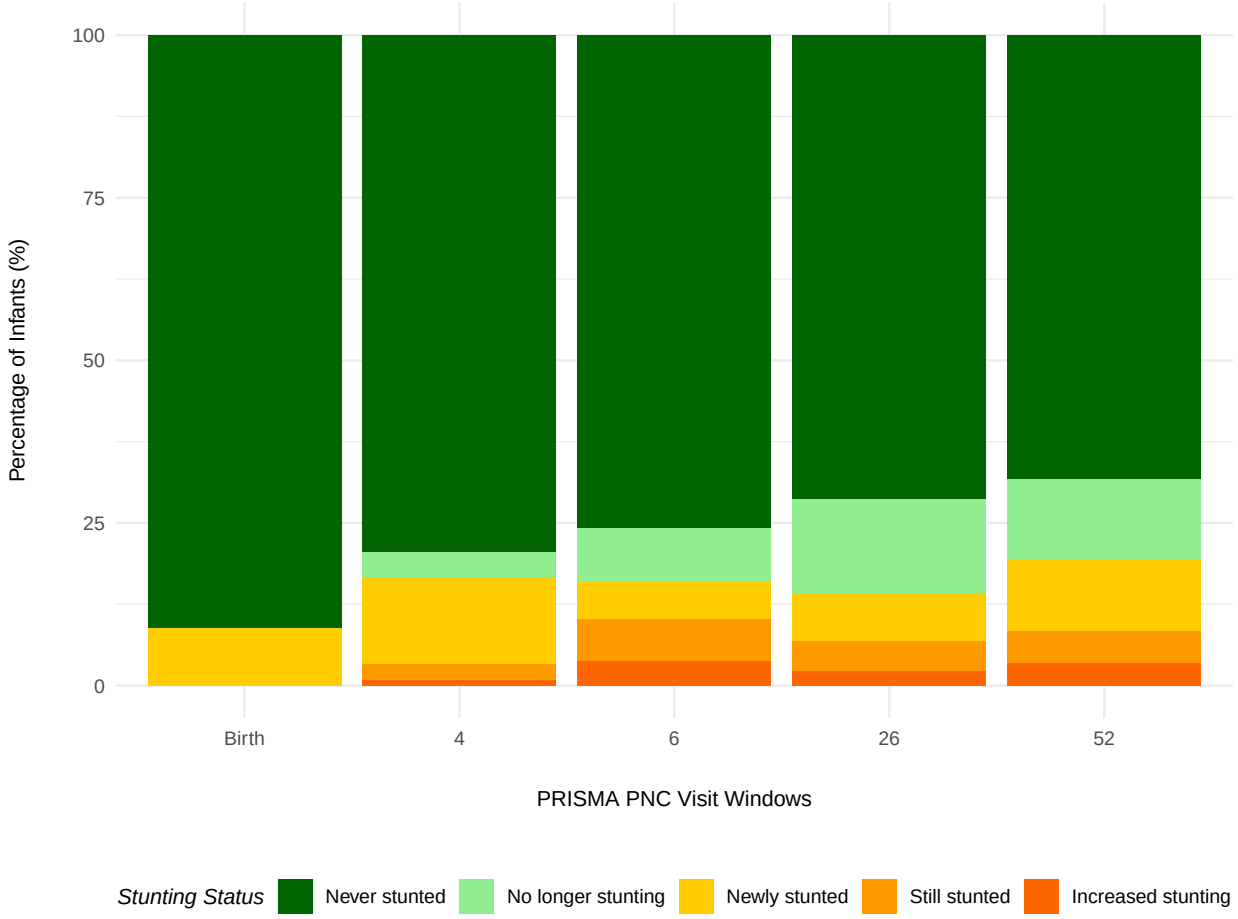


Figure 6: cummulative change in stunting across PRISMA PNC visit windows

Definitions of stunting status categories

1. **Never Stunted**: Children who have never been stunted at any recorded time point .
2. **No Longer Stunting**: Children who were previously stunted but are not stunted in the current time point.
3. **Newly Stunted**: Children who were not stunted in the previous time point but are stunted in the current time point.
4. **Still Stunted**: Children who remain stunted across consecutive time points.
5. **Increased Stunting**: Children who were stunted in the previous time point and have transitioned to severe stunting.
6. **Stunting Relapse**: Children who were not stunted previously, then stunted, recovered, and are stunted again.



13. Exclusive Breastfeeding Analysis

Exclusive breastfeeding (EBF) at a given postnatal timepoint is defined cumulatively.

An infant is classified as **exclusively breastfed** at a given timepoint if the caregiver reported exclusive breastfeeding at that visit and at all prior postnatal visits, with no reported intake of any non-breast-milk foods or liquids at any earlier visit.

If an infant is reported as receiving any non-breast-milk food or liquid at any visit, the infant is considered not exclusively breastfed from that time onward and remains classified as non-EBF at all subsequent timepoints.

For the following tables and descriptive analysis, exclusive breastfeeding (EBF) patterns were examined across the early postnatal period, with assessments conducted at both the 6-week and 6-month postnatal visits to capture changes in infant feeding practices over time. The analysis evaluate data availability and measurement completeness, estimate the proportion of infants exclusively breastfed, and explore variation in EBF prevalence across combined size-for-gestational-age and gestational-age (Small Vulnerable Newborn - SVN) groupings.

13a. Infant exclusive breastfeeding status at 6weeks

Infant exclusive breastfeeding (EBF) status was assessed at the 6-week postnatal care visit to describe feeding practices during early infancy. The following tables summarize the completeness of EBF data, the prevalence of exclusive breastfeeding overall and across infant size-for-gestational-age and gestational-age (SVN) categories, and caregiver-reported feeding behaviors among infants who were not exclusively breastfed.

For infants not exclusively breastfed at 6 weeks, additional summaries characterize reported reasons for non-exclusive breastfeeding and the types of non-breastmilk foods or liquids introduced. These caregiver-reported measures allow multiple responses; therefore, percentages may sum to more than 100 percent.

Together, these analyses provide an integrated overview of early infant feeding practices and contextual factors influencing exclusive breastfeeding at the 6-week postnatal timepoint.

Table 13a (i). Infant exclusive breastfeeding completion status at 6weeks

Table 13a(i). EBF data completeness at 6weeks

	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Denominator [a]	2357	2202	1906	2368	5060	1518
Invalid EBF	100 (4.2)	55 (2.5)	38 (2)	71 (3)	147 (2.9)	20 (1.3)
Missing SVN Group	80 (3.4)	9 (0.4)	14 (0.7)	46 (1.9)	476 (9.4)	89 (5.9)
Eligible denominator [b]	2201	2143	1861	2254	4476	1411

[a] Denominator includes all infants with non-missing `VISIT_DATE_PNC6` in `INF_EBF_WIDE`.

[b] Eligible denominator excludes invalid EBF values, missing SVN grouping, and (for non-EBF infants) missing foods/reason fields.

Abbreviations: EBF, exclusive breastfeeding; SVN, size-for-gestational-age and gestational age category.

Table 13a (ii). Infant exclusive breastfeeding prevalence overall and by SVN group at 6weeks

Table 13a(ii). Exclusive breastfeeding at 6weeks by PRISMA Site and SVN group						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Overall						
Denominator (N)	2201	2143	1861	2254	4476	1411
EBF n(%)	2076 (94.3)	1744 (81.4)	573 (30.8)	2127 (94.4)	3603 (80.5)	1191 (84.4)
Term AGA						
Denominator (N)	1461	1529	1028	1689	2441	930
EBF n(%)	1386 (94.9)	1280 (83.7)	341 (33.2)	1617 (95.7)	2013 (82.5)	792 (85.2)
Term SGA						
Denominator (N)	603	441	555	380	1146	315
EBF n(%)	573 (95.0)	354 (80.3)	166 (29.9)	356 (93.7)	929 (81.1)	281 (89.2)
Preterm AGA						
Denominator (N)	104	137	212	159	722	113
EBF n(%)	92 (88.5)	94 (68.6)	52 (24.5)	134 (84.3)	546 (75.6)	85 (75.2)
Preterm SGA						
Denominator (N)	33	36	66	26	167	53
EBF n(%)	25 (75.8)	16 (44.4)	14 (21.2)	20 (76.9)	115 (68.9)	33 (62.3)

Notes: EBF n(%) is the number of infants exclusively breastfed (n) and percent (%) among the denominator (N) within each site and SVN group.

SVN groups: Term AGA = term or postterm infants (≥37 weeks) who are appropriate-for-gestational-age (10th to <90th percentile) or large-for-gestational-age (≥90th percentile); Term SGA = term or postterm infants (≥37 weeks) who are small-for-gestational-age (<10th percentile); Preterm AGA = preterm infants (<37 weeks) who are appropriate-for-gestational-age or large-for-gestational-age; Preterm SGA = preterm infants (<37 weeks) who are small-for-gestational-age.

Abbreviations: EBF, exclusive breastfeeding; SVN, combined size-for-gestational-age and gestational age category.

Table 13a (iii). Reasons why infant are not exclusively breastfed as reported by caregiver at 6weeks

Table 13a(iii). Reasons for not exclusive breastfeeding at 6weeks, by site						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Denominator	125	399	1288	127	873	220
Congenital malformation n(%)	2 (1.6)	1 (0.3)	1 (0.1)	2 (1.6)	5 (0.6)	1 (0.5)
Infant greater than 6 months n(%)	0 (0)	2 (0.5)	0 (0)	8 (6.3)	8 (0.9)	0 (0)
Infant illness n(%)	9 (7.2)	176 (44.1)	102 (7.9)	22 (17.3)	52 (6.0)	10 (4.5)
Interference from family or health provider n(%)	20 (16.0)	153 (38.3)	121 (9.4)	0 (0)	126 (14.4)	2 (0.9)
Latching problem n(%)	24 (19.2)	12 (3.0)	18 (1.4)	2 (1.6)	15 (1.7)	3 (1.4)
Maternal death or absence n(%)	0 (0)	0 (0)	3 (0.2)	0 (0)	8 (0.9)	2 (0.9)
Maternal illness n(%)	10 (8.0)	24 (6.0)	97 (7.5)	3 (2.4)	41 (4.7)	13 (5.9)
Maternal preference n(%)	79 (63.2)	109 (27.3)	892 (69.3)	44 (34.6)	673 (77.1)	103 (46.8)
PMTCT n(%)	0 (0)	1 (0.3)	2 (0.2)	25 (19.7)	0 (0)	12 (5.5)
Other reasons n(%)	14 (11.2)	1 (0.3)	632 (49.1)	35 (27.6)	50 (5.7)	141 (64.1)

Abbreviations: EBF, exclusive breastfeeding; PNC6, postnatal care visit at 6 weeks

Notes: The denominator (N) represents the number of eligible infants who were not exclusively breastfed at 6weeks (EBF_PNC6 = 0) within each site. Reasons are multiple-response; infants may contribute more than one reason, so percentages may sum to >100. Codes 55 (missing information) and 77 (infant is breastfed) are excluded.

Table 13a (iv). Other foods given to infants not exclusively breastfed as reported by caregiver at 6weeks

Table 13a(iv). Non-EBF foods at 6weeks, overall and by SVN group					
	Overall	Preterm AGA	Preterm SGA	Term AGA	Term SGA
Denominator	3032	444	158	1649	781
Animal milk n(%)	849 (28.0)	90 (20.3)	34 (21.5)	475 (28.8)	250 (32.0)
Breastmilk n(%)	2216 (73.1)	312 (70.3)	116 (73.4)	1199 (72.7)	589 (75.4)
Formula n(%)	1837 (60.6)	320 (72.1)	119 (75.3)	923 (56.0)	475 (60.8)
Gruel/porridge n(%)	89 (2.9)	6 (1.4)	2 (1.3)	58 (3.5)	23 (2.9)
Honey n(%)	25 (0.8)	7 (1.6)	0 (0)	12 (0.7)	6 (0.8)
Other n(%)	737 (24.3)	92 (20.7)	24 (15.2)	426 (25.8)	195 (25.0)
Water n(%)	637 (21.0)	69 (15.5)	30 (19.0)	366 (22.2)	172 (22.0)

Notes: Table includes eligible infants who were not exclusively breastfed at six weeks. Percentages use the number of not exclusively breastfed infants within each SVN group as the denominator. Foods are multiple-response; infants may report more than one food, so percentages may sum to >100. Infants exclusively breastfed (code 0) and infants with missing food information (code 55) are excluded from food categories.

Abbreviations: EBF, exclusive breastfeeding; SVN, size-for-gestational-age and gestational age category.

13b. Infant exclusive breastfeeding status at 26weeks

Infant exclusive breastfeeding (EBF) status was assessed at the 6-week postnatal care visit to describe feeding practices during early infancy. The following tables summarize the completeness of EBF data, the prevalence of exclusive breastfeeding overall and across infant size-for-gestational-age and gestational-age (SVN) categories, and caregiver-reported feeding behaviors among infants who were not exclusively breastfed.

For infants not exclusively breastfed at 6 weeks, additional summaries characterize reported reasons for non-exclusive breastfeeding and the types of non-breastmilk foods or liquids introduced. These caregiver-reported measures allow multiple responses; therefore, percentages may sum to more than 100 percent.

Together, these analyses provide an integrated overview of early infant feeding practices and contextual factors influencing exclusive breastfeeding at the 6-week postnatal timepoint.

Table 13b (i). Infant exclusive breastfeeding completion status at 26weeks

Table 13b(i). EBF data completeness at 26weeks

	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Denominator [a]	2357	2202	1906	2370	5060	1518
Invalid EBF	66 (2.8)	29 (1.3)	39 (2)	70 (3)	122 (2.4)	35 (2.3)
Missing SVN Group	80 (3.4)	9 (0.4)	14 (0.7)	47 (2)	476 (9.4)	89 (5.9)
Eligible denominator [b]	2228	2166	1860	2257	4487	1395

[a] Denominator includes all infants with non-missing `VISIT_DATE_PNC26` in `INF_EBF_WIDE`.

[b] Eligible denominator excludes invalid EBF values, missing SVN grouping, and (for non-EBF infants) missing foods/reason fields.

Abbreviations: EBF, exclusive breastfeeding; SVN, size-for-gestational-age and gestational age category.

Table 13b (ii). Infant exclusive breastfeeding prevalence overall and by SVN group at 26weeks

Table 13b(ii). Exclusive breastfeeding at 26weeks by PRISMA Site and SVN group						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Overall						
Denominator (N)	2228	2166	1860	2257	4487	1395
EBF n(%)	1205 (54.1)	620 (28.6)	58 (3.1)	1044 (46.3)	1481 (33.0)	253 (18.1)
Term AGA						
Denominator (N)	1479	1542	1024	1691	2448	919
EBF n(%)	824 (55.7)	441 (28.6)	30 (2.9)	806 (47.7)	820 (33.5)	168 (18.3)
Term SGA						
Denominator (N)	611	446	557	381	1147	312
EBF n(%)	315 (51.6)	132 (29.6)	24 (4.3)	172 (45.1)	374 (32.6)	47 (15.1)
Preterm AGA						
Denominator (N)	105	141	213	159	725	112
EBF n(%)	49 (46.7)	40 (28.4)	3 (1.4)	54 (34.0)	239 (33.0)	27 (24.1)
Preterm SGA						
Denominator (N)	33	37	66	26	167	52
EBF n(%)	17 (51.5)	7 (18.9)	1 (1.5)	12 (46.2)	48 (28.7)	11 (21.2)

Notes: EBF n(%) is the number of infants exclusively breastfed (n) and percent (%) among the denominator (N) within each site and SVN group.

SVN groups: Term AGA = term or postterm infants (≥37 weeks) who are appropriate-for-gestational-age (10th to <90th percentile) or large-for-gestational-age (≥90th percentile); Term SGA = term or postterm infants (≥37 weeks) who are small-for-gestational-age (<10th percentile); Preterm AGA = preterm infants (<37 weeks) who are appropriate-for-gestational-age or large-for-gestational-age; Preterm SGA = preterm infants (<37 weeks) who are small-for-gestational-age.

Abbreviations: EBF, exclusive breastfeeding; SVN, combined size-for-gestational-age and gestational age category.

Table 13b (iii). Reasons why infant are not exclusively breastfed as reported by caregiver at 26weeks

Table 13b(iii). Reasons for not exclusive breastfeeding at 26weeks, by site						
	Ghana	India-CMC	India-SAS	Kenya	Pakistan	Zambia
Denominator	1023	1546	1802	1213	3006	1142
Congenital malformation n(%)	2 (0.2)	1 (0.1)	1 (0.1)	2 (0.2)	5 (0.2)	1 (0.1)
Infant greater than 6 months n(%)	897 (87.7)	1361 (88.0)	1380 (76.6)	1074 (88.5)	2591 (86.2)	0 (0)
Infant illness n(%)	10 (1.0)	177 (11.4)	105 (5.8)	23 (1.9)	58 (1.9)	10 (0.9)
Interference from family or health provider n(%)	40 (3.9)	158 (10.2)	122 (6.8)	1 (0.1)	134 (4.5)	4 (0.4)
Latching problem n(%)	24 (2.3)	12 (0.8)	20 (1.1)	2 (0.2)	16 (0.5)	3 (0.3)
Maternal death or absence n(%)	1 (0.1)	0 (0)	3 (0.2)	0 (0)	11 (0.4)	2 (0.2)
Maternal illness n(%)	15 (1.5)	25 (1.6)	99 (5.5)	3 (0.2)	44 (1.5)	14 (1.2)
Maternal preference n(%)	242 (23.7)	164 (10.6)	1003 (55.7)	173 (14.3)	691 (23.0)	544 (47.6)
PMTCT n(%)	0 (0)	1 (0.1)	2 (0.1)	31 (2.6)	0 (0)	15 (1.3)
Other reasons n(%)	31 (3.0)	1 (0.1)	642 (35.6)	38 (3.1)	57 (1.9)	804 (70.4)

Abbreviations: EBF, exclusive breastfeeding; PNC26, postnatal care visit at 6 months.

Notes: The denominator (N) represents the number of eligible infants who were not exclusively breastfed at 26weeks (EBF_PNC26 = 0) within each site. Reasons are multiple-response; infants may contribute more than one reason, so percentages may sum to >100. Codes 55 (missing information) and 77 (infant is breastfed) are excluded.

Table 13b (iv). Other foods given to infants not exclusively breastfed as reported by caregiver at 26weeks

Table 13b(iv). Non-EBF foods at 226weeks, overall and by SVN group					
	Overall	Preterm AGA	Preterm SGA	Term AGA	Term SGA
Denominator	9732	1043	285	6014	2390
Animal milk n(%)	3508 (36.0)	341 (32.7)	93 (32.6)	2197 (36.5)	877 (36.7)
Breastmilk n(%)	8381 (86.1)	850 (81.5)	241 (84.6)	5191 (86.3)	2099 (87.8)
Formula n(%)	2820 (29.0)	407 (39.0)	143 (50.2)	1540 (25.6)	730 (30.5)
Gruel/porridge n(%)	6752 (69.4)	599 (57.4)	156 (54.7)	4376 (72.8)	1621 (67.8)
Honey n(%)	2244 (23.1)	165 (15.8)	43 (15.1)	1486 (24.7)	550 (23.0)
Other n(%)	3752 (38.6)	390 (37.4)	100 (35.1)	2326 (38.7)	936 (39.2)
Water n(%)	7709 (79.2)	764 (73.3)	196 (68.8)	4839 (80.5)	1910 (79.9)

Notes: Table includes eligible infants who were not exclusively breastfed at 26 weeks. Percentages use the number of not exclusively breastfed infants within each SVN group as the denominator. Foods are multiple-response; infants may report more than one food, so percentages may sum to >100. Infants exclusively breastfed (code 0) and infants with missing food information (code 55) are excluded from food categories.

Abbreviations: EBF, exclusive breastfeeding; SVN, size-for-gestational-age and gestational age category.